ALLSEIS-1CHR Node for High Density Seismic

Built for High Resolution, High Efficiency, and High Reliability Seismic Acquisition with Less Operational Cost and Environmental Impact

General Features

Sensor	Internal 5Hz high sensitivity Geophone Internal 10Hz high sensitivity geophone Optional KCK version for external string
Digitalization	32-bit Δ-Σ ADC
Storage Capacity	16GB (32GB optional)
Battery Life	Internal rechargeable Li-Ion battery (52WH) supports 800 hours continuous recording
Timing Accuracy	+/- 10μs (continuous GNSS disciplined) +/- 1ms (within 1 hour without GNSS signal)
Automated Test	Internal testing signal generator for function and performance self-inspection
Working Mode	Autonomy with wireless on-site QC inspection, support UAV patrol
LED Status Indicator	Nodal health, GNSS synchronization status, Wireless data harvesting status
Data Harvest	Charging/Data download rack
Infield Management	Industrial handheld android tablet
Weight	750g （excl. spike），800g（incl. spike）
External Size	9.8L x 10.8W x 11H cm（excl. spike）
Operating Temperature	-40°C ~ +70°C

Acquisition Specifications

ADC Resolution	32-bit	
Sample Interval	0.25ms、0.5ms、1ms、2ms、4ms	
Preamplifier Gain	x1、x2、x4、x8、x16、x32、x64 （0dB、6dB、12dB、18dB、24dB、30dB、36dB）	
Input Impedance	40KΩ || 22nF (5Hz geophone) 20KΩ || 22nF (10Hz geophone)	
Gain Accuracy	0.1%	
Full Scale Input Signal	±2.5Vp-p@x1gain ±156mVp-p@x16gain	±625mVp-p@x4gain ±39mVp-p@x64gain
Instantaneous DR @4ms（typical value）	130dB @ x1 gain 123dB @ x16 gain	128dB @ x4 gain 114dB @ x64 gain
Full Scale DR	150dB	
Equivalent Input Noise @4ms（typical value）	0.50μV @ x1 gain 0.08μV @ x16 gain	0.15μV @ x4 gain 0.07μV @ x64 gain
Harmonic Distortion	-120dB @ 31.5Hz	
Frequency Response @0.25ms	0Hz ~ 1652Hz (-3dB)	
Anti-alias Filter	Linear or minimum phase filter, 82.6% of Nyquist -3dB bandwidth	
DC-blocking filter	0.1 ~ 10Hz	
Stopband Attenuation	>130dB @Nyquist	
Common Mode Rejection	>120dB	

Product Highlight

- All-in-one seismic acquisition nodal system with built-in 5Hz (5Hz~150Hz) or 10Hz (10Hz~240Hz) high sensitivity geophone, optional KCK version for external geophone string;
- 32-bit Δ-Σ ADC converter for high resolution, high dynamic range seismic signal recording, with configurable sample rate and preamplifier gain;
- High sensitive GNSS module equipped with active high-gain patch antenna supports GPS, Beidou, and GLONASS;
- Low power intermittent GNSS clock synchronization strategy with timing accuracy better than ± 10us, support continuous recording for up to 800 hours (deep burial not recommended);
- Configurable continuous GNSS clock synchronization strategy, with timing accuracy less than ±1us, support deep supports deep planting applications (typically 1m buried depth in sand) ;
- Built-in test signal generator supports on-site full signal chain function and performance self-inspection and diagnosis, including system noise, real-time dynamic range, geophone impedance, natural frequency, sensitivity, damping, distortion, etc.;
- BLE wireless module together with visual LEDs for onsite configuration and proximity. Ground communication distance ≥20m, air communication distance ≥100m, supports vehicle or UAV patrol;
- Industrial Android tablet supports node status verification after deployment, including battery autonomy, environmental temperature, tilting, ambient noise, real-time seismic waveform, etc.;
- P&R(Place & Record) mode for rapid and convenient infield deployment, supporting automatic position matching with line and point number;
- Waterproof(IP68), flame retardant (5VA), ultraviolet aging (FI);
- Built-in electronic tag (RFID) for efficient asset management;
- Compact volume (9.8L x 10.8W x 11H cm) and light weight (750g), support acquisition under -55°C.

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ALLSEIS-1CHR Node for High Density Seismic

中科深源

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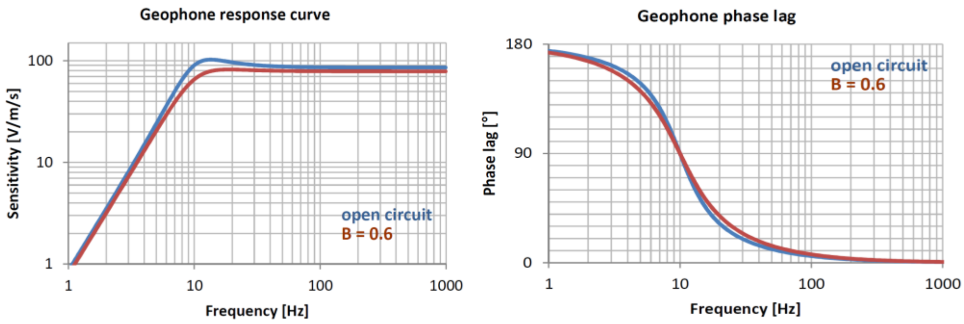
10Hz High Sensitivity Geophone

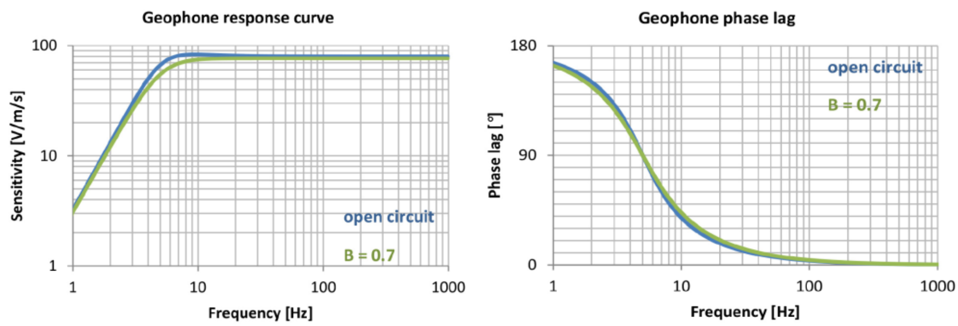
Natural Frequency	10Hz
Sensitivity	85.8V/m/s
Spurious	>240Hz
Distortion	≤0.1%
Coil Resistance	1800Ω
Tilt	±20°



10Hz High Sensitivity Geophone Response Curve and Phase Lag

5Hz High Sensitivity Geophone

Natural Frequency	5Hz
Sensitivity	80V/m/s
Spurious	>150Hz
Distortion	≤0.1%
Coil Resistance	1850Ω
Tilt	±10°



5Hz High Sensitivity Geophone Response Curve and Phase Lag

Charging Rack

Slots	32
Charging Time	3.5 hours
Charging Power	700W
Operating Temperature	0°C to +40°C
Size	194x64x38cm

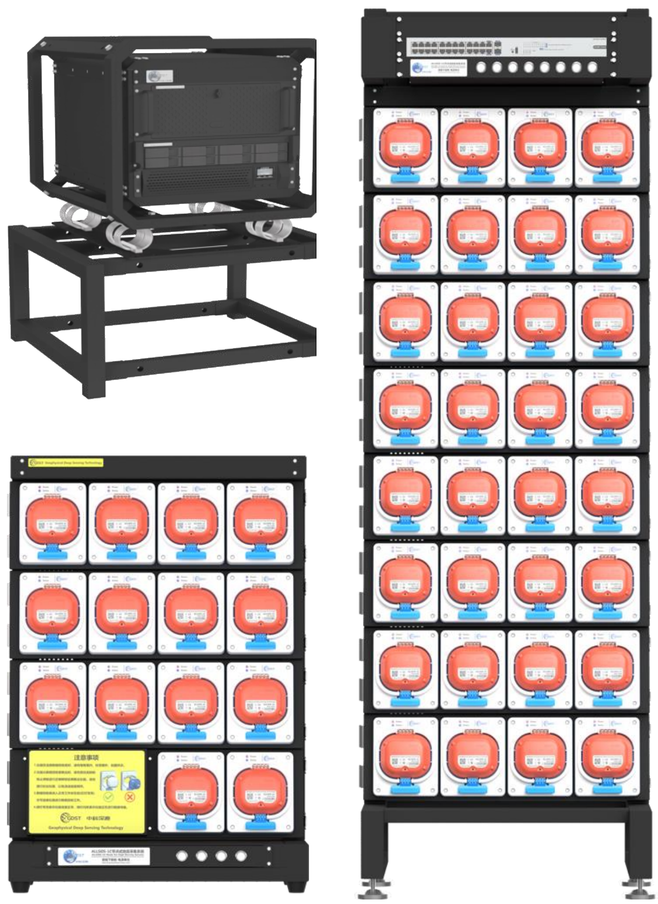
Data Download Rack

Slots	32
Unit Download Rate	0.6GB/min.
Full Download Rate	18GB/min.
Interface	10G SFP+
Operating Temperature	0°C to +40°C
Size	194x64x38cm

Desktop Data download/Charging Rack

Slots	14
Unit Download Rate	0.6GB/min.
Full Download Rate	8GB/min.
Interface	2.5G Ethernet
Charging Time	3.5 hours
Charging Power	350W
Working Temperature	0°C to +40°C
Size	94x64x38cm

ALLSEIS-1CHR:
Standard & KCK Version



Supporting Equipment:
Data Management Server, Charging/Data Downloading Rack

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